

MOAZ MOHAMMED HUSAIN

AI/ML Engineer

Mumbai, Maharashtra

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SKILLS

- **Programming Languages:** Python, R, SQL, C++, Java, Shell Scripting
- **Machine Learning:** ML algorithms, Model development, image classification, speech recognition, time-series forecasting
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, GAN, DNN, GNN, Transformer Models, Multimodal Models
- **Libraries:** Scikit-learn, Numpy, Pandas, Xgboost, TensorFlow, PyTorch, Keras, CUDA, YOLO
- **Computer Vision:** OpenCV, Pillow, Stable Diffusion, Dreambooth, LoRA, GANs, Video Diffusion Models
- **Natural Language Processing (NLP):** BERT, LLMs, RAG, Text Classification
- **Generative AI:** LangChain, GPT, Cohere, Mistral, Llama 3, Weaviate
- **Large Language Models (LLMs):** Training and fine-tuning, Multimodal modeling (audio, text, image), Prompt engineering
- **Data Analysis:** Exploratory Data Analysis (EDA), Predictive Analytics, Time Series Analysis, Risk Modeling, Sentiment Analysis
- **Data Engineering:** ETL pipelines, Synthetic data generation, Data augmentation, Large-scale dataset curation
- **Web Framework:** Flask, Django, FastAPI, Streamlit, REST API
- **Cloud Platforms and Containerization:** AWS, GCP, Azure, Databricks, Docker, Kubernetes
- **Software Engineering:** Git, Linux, Bash scripting, Agile methodologies, Unit testing, Code optimization
- **Optimization:** Hyperparameter tuning, performance benchmarking, memory management, resource allocation, model scalability.
- **DevOps & MLOps:** CI/CD Pipelines, MLflow, Version Control (Git, GitHub), Automation Scripts
- **Agile Development:** Scrum, Kanban

WORK EXPERIENCE

AI Engineer

April 2024 - January 2025

Outlier

- Led Reinforcement Learning from Human Feedback (RLHF) and prompt engineering projects, improving AI model accuracy by 30% and reducing training time.
- Streamlined model training workflows, enhancing generative AI efficiency by 25%, accelerating time-to-insight.
- Optimized AI model outputs through prompt engineering, improving alignment with user intent.
- Improved data quality through prompt design, achieving a 95% satisfaction rate for insights used in claims processing.

AI Engineer

July 2022 - December 2023

AI Hub Durham College

- Developed an NLP-based financial case categorization system, processed and analyzed unstructured text data to classify cases into predefined categories, improving classification accuracy to 88% over 100K+ documents.
- Engineered a YOLO-based system for vehicle and license plate identification, achieving 92% accuracy on license plate recognition at 200+ FPS.
- Designed and deployed REST API for artist data insights, integrating machine learning models into production systems to deliver actionable analytics.
- Designed MLOps workflows with MLflow and CI/CD, automating model deployment and reducing downtime by 40%.
- Conducted deep learning research on Transformer model efficiency, optimizing attention mechanisms to reduce memory overhead by 15%.

Data Scientist

January 2019 - December 2022

Fiverr

- Developed scalable backend APIs for AI applications using FastAPI and Flask, reducing inference latency by 30%.
- Built automated text classification pipelines using LLMs and XGBoost, achieving 95% classification accuracy on client datasets.
- Designed and implemented large-scale ETL workflows, processing 100K+ documents per month for automated data extraction.
- Integrated GAN-based data augmentation techniques, improving model generalization and boosting dataset diversity by 40%.
- Created end-to-end CI/CD pipelines for AI models, accelerating model deployment cycles by 50%.

Data Scientist

September 2020 - December 2020

Fellowshipai

- Developed a CNN-based computer vision model for food image classification, reaching 92% accuracy in detecting pizza doneness stages.
- Trained GAN models for synthetic image generation, improving visual realism and dataset diversity by 30%.
- Researched Transformer model architectures, identifying optimizations that reduced inference latency by 25%.

PROJECTS

Credit Card Fraud Detection | [Link](#)

- Engineered a machine learning pipeline to detect credit card fraud in imbalanced datasets, improving detection accuracy by 92%.
- Developed a Streamlit-based web application for real-time fraud prediction, ensuring accessible user interaction with predictive models.

Wine Quality Prediction Using Databricks | [Link](#)

- Developed a data pipeline (using Python) to integrate with Databricks for scalable data processing and model training of a wine quality prediction system achieving 90% accuracy.
- Implemented data preprocessing techniques and utilized MLflow for experiment tracking, model management, and hyperparameter tuning.
- Integrated the project with Databricks for scalable data processing and model training.

Image Animation with Generative Adversarial Networks (GANs) | [Link](#)

- Designed a GAN model using PyTorch to transform images into anime-style visuals, showcasing expertise in generative modeling and advanced pre-processing techniques.
- Achieved enhanced image quality and diversity through hyperparameter optimization, demonstrating proficiency in model tuning and experimentation.

Diabetes FAQ Chatbot with GPT-3 | [Link](#)

- Leveraged LLM (GPT-3) to design a RAG system for building a chatbot to answer common diabetes-related queries, retrieving knowledge from a vector database, automating responses for common customer inquiries, and enhancing overall user satisfaction.

Podcast Q&A Assistant with Multimodal RAG | [Link](#)

- Built an end-to-end AI system (Whisper + Weaviate + Cohere) that transcribes podcasts and answers questions, deployed via Streamlit.
- Optimized semantic search through custom chunking and hybrid retrieval (vector + keyword), reducing latency by 40% compared to baseline approaches.

Flower Classification with Azure ML | [Link](#)

- Developed a flower classification model using Azure AI, achieving 85% accuracy and deploying the model as a REST API on Azure App Service.
- Demonstrated proficiency in Python, machine learning, Azure AI suite, and cloud platforms by building a complete ML pipeline from data ingestion to model deployment and maintenance.

EDUCATION

Post-Graduate in Artificial Intelligence Durham College	Oshawa, Ontario, Canada
Bachelor of Engineering in Electronics Engineering University of Mumbai	Mumbai, Maharashtra, India

CERTIFICATION

- [Data Science](#), IBM
- [Machine Learning](#), IBM
- [Natural Language Processing](#), Deeplearning.ai
- [Data Engineering](#), IBM